

PAVIT AGRAWAL

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PROFESSIONAL SUMMARY

Full-stack software engineer with demonstrated expertise in building scalable, production-grade applications spanning backend services, frontend interfaces, and AI-driven systems. Proficient in modern architectures (microservices, serverless), cloud platforms, and advanced technologies including agentic AI, multimodal systems, and spatial intelligence. Proven ability to design and deliver end-to-end solutions from concept to deployment. Strong problem-solving skills with hands-on experience in fraud detection, compliance systems, and high-performance data processing. Passionate about clean code, system design, and leveraging cutting-edge technologies to solve complex problems.

EDUCATION

Bachelor of Technology, Computer and Communication Engineering

Manipal Institute of Technology, Manipal, Karnataka | Jul 2023 – May 2027

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Discrete Mathematics, Linear Algebra, Calculus, Machine Learning, Software Engineering, System Design

Senior Secondary (PCM)

Delhi Public School, Indirapuram | Mar 2023

TECHNICAL EXPERTISE

CORE COMPETENCIES

Full-Stack Development • System Architecture & Design • API Development & Integration • Scalable Backend Services • Modern Frontend Development • Cloud Infrastructure • AI/LLM Integration • Agentic Workflows • Performance Optimization • Security & Authentication • Problem Solving • Critical Thinking • Fast Learner • Team Collaborator • Communication Skills • Self-Driven & Proactive • Code Quality Focus • Attention to Detail

Languages | Python, C++, Java, JavaScript, TypeScript, SQL, PL/SQL, Bash

Backend Development | FastAPI, Node.js, Express, Django, REST APIs, WebSockets, Microservices, SOLID Principles

Frontend Development | React, Next.js 15, React 19, HTML5, CSS3, Tailwind CSS, TypeScript, State Management

Databases & ORM | PostgreSQL, MySQL, MongoDB, SQLAlchemy, Pydantic, PostGIS, Query Optimization

AI/ML & LLMs | Agentic AI, LangGraph, Hybrid GraphRAG, RAG, MCP, Gemini API, Claude API, NLP, Deep Learning, scikit-learn, Prompt Engineering, LLM Security

Cloud & DevOps | AWS (Certified), Docker, Vercel, Render, Supabase, CI/CD, Git, GitHub Actions

Data & Analytics | Power BI, Tableau, EDA, Statistical Modeling, GIS/LiDAR, Geospatial Intelligence

Specialized Domains | Fraud Detection Systems, Compliance Auditing, Multimodal AI, Spatial Intelligence, Real-time Processing

PROFESSIONAL EXPERIENCE

Control Assurance Intern

Deloitte, Audit & Assurance • Delhi NCR, India | May 2026 – Jul 2026

Collaborated on enterprise IT audit engagements and control assessments. Assessed Digital Personal Data Protection (DPDP) Act compliance for industrial clients; verified and validated cybersecurity controls including password policies, access-control mechanisms, and infrastructure configurations. Documented control maturity, gap analysis, and remediation strategies.

Accomplishments:

Conducted comprehensive control testing and documented security/compliance gaps across enterprise systems
Performed technical analysis of access controls, authentication mechanisms, and data protection implementations
Prepared detailed audit reports with evidence-based remediation recommendations
Collaborated with senior auditors and IT teams on multi-client engagements

Data Analyst Intern

TransOrg Analytics • Gurugram, India | May 2025 – Jul 2025

Contributed to ML-driven fraud detection system development and production integration. Analyzed product-level sales and margin impact for retail optimization initiatives. Built data processing pipelines, EDA frameworks, and BI dashboards using Python, SQL, and modern data stack tools.

Accomplishments:

Supported development of production deep-learning fraud detection engine with cross-functional teams
Built SQL pipelines and Python data processing modules for feature engineering and validation
Developed Power BI and Tableau dashboards for stakeholder reporting and business intelligence
Mastered full analytics stack: SQL optimization, Python (pandas, NumPy), BI tools, and statistical modeling

FEATURED PROJECTS

Civitas: Multimodal Civic Incident Intelligence Platform

2026 • Full-Stack: FastAPI • Next.js/React • LangGraph • PostgreSQL • PostGIS • Docker

End-to-end platform transforming unstructured citizen reports (images, video, text, GPS) into verified, prioritized municipal work orders. Demonstrates full-stack expertise: intelligent multimodal AI processing on backend, geospatial querying, and responsive real-time frontend.

Architecture & Technical Highlights:

Backend: FastAPI microservices with async task processing and multimodal AI model integration

Geospatial: PostGIS queries for location-based clustering, priority ranking, and resource optimization

Frontend: React/Next.js interactive maps with real-time status updates and incident dashboard

DevOps: Docker containerization, scalable deployment architecture

AI/ML: Vision models for image classification, NLP for text analysis, fusion algorithm for prioritization

Mnemos: Industrial AI Knowledge Intelligence Platform

2025 • Backend Architecture: Python • FastAPI • LangGraph • Hybrid GraphRAG • MCP

Advanced agentic platform combining hybrid retrieval (vector embeddings + knowledge graphs + structured data) for enterprise asset intelligence. Implements human-in-the-loop workflows, evidence-grounded reasoning, and multi-agent orchestration.

Architecture & Technical Highlights:

Agent Design: LangGraph-based orchestration for multi-agent workflows and decision-making pipelines

Retrieval: Hybrid GraphRAG combining vector search (semantic), graph queries (relationships), and SQL (structured)

Integration: MCP (Model Context Protocol) for tool-agnostic capability orchestration

Compliance: Human approval workflows, audit trails, evidence traceability for regulatory requirements

Backend: FastAPI with async processing, dependency injection, comprehensive error handling

Live Demo: <https://mnemos-demo.vercel.app>

GitHub Repository: <https://github.com/pavitagrawal/mnemos-platform>

A-DAP-T: AI-Agent Safety Scanner

2025 • Backend: Python • FastAPI • Security Analysis • Vulnerability Detection

Deployment safety scanner providing pre-production security analysis for AI-agent systems. Identifies prompt-injection vulnerabilities, credential exposure, unsafe permission configurations, and deployment risks.

Architecture & Technical Highlights:

Security Analysis: Prompt-injection detection via adversarial testing and pattern matching

Secrets Detection: Scanning configs, environment files, and code comments for credential exposure

Permission Analysis: Tool permission matrix validation and capability risk assessment

API Design: RESTful endpoints with comprehensive report generation and remediation guidance

GitHub Repository: <https://github.com/pavitagrawal/adapt-safety-scanner>

Hydro-DTM System: Terrain Classification & Digital Terrain Modeling

2025 • Backend/Data Pipeline: Python • GIS/LiDAR • Machine Learning • Scikit-learn

End-to-end terrain classification and digital terrain modeling pipeline. Demonstrates expertise in geospatial data processing, feature engineering, ML model optimization, and production-grade system design. Achieved 95%+ classification accuracy and 90% predictive accuracy.

Recognition: Finalist – National Geo AI Hackathon, IIT Bombay Techfest 2025

Architecture & Technical Highlights:

Data Pipeline: LiDAR point cloud ingestion, preprocessing, and feature extraction

ML Models: Ensemble approach (Random Forest, Gradient Boosting) with hyperparameter optimization

Geospatial Processing: GIS raster/vector operations for terrain model generation

Validation: Cross-validation, ROC-AUC analysis, confusion matrices, performance benchmarking

GitHub Repository: <https://github.com/pavitagrawal/hydro-dtm-system>

Daedalus: AI Career Navigator Platform

2025 • Backend: FastAPI • Pydantic • SQLAlchemy • PostgreSQL • REST APIs

RESTful backend platform providing AI-driven career assessment and personalized recommendations. Demonstrates clean architecture, database design, API best practices, and algorithmic problem-solving.

Architecture & Technical Highlights:

API Design: RESTful endpoints for career assessment, skill analysis, and recommendations

Database: PostgreSQL schema design with normalized relationships and query optimization

Validation: Pydantic models for request/response validation with comprehensive error handling

Algorithms: Deterministic scoring algorithms for career-fit analysis and skill-gap identification

GitHub Repository: <https://github.com/pavitagrawal/daedalus-career-navigator>

Bharat Finance Platform

2025 • Frontend: Next.js 15 • React 19 • Cloud: AWS • Styling: Tailwind CSS

Production-ready FinTech platform featuring multilingual interface with real-time financial data visualizations. Built for AWS ImpactX Hackathon at IIT Bombay Techfest 2025. Demonstrates modern frontend architecture, performance optimization, and cloud deployment.

Architecture & Technical Highlights:

Frontend: Next.js 15 server components for optimal performance and SEO

UI/UX: React 19 component architecture with responsive design and accessibility

Data Visualization: Interactive charts and dashboards with real-time data updates

i18n: Multilingual support for regional financial terminology and user experience

Cloud: Deployed as scalable, production-ready application on AWS infrastructure

Live Application: <https://bharat-finance-platform.vercel.app>

GitHub Repository: <https://github.com/pavitagrawal/bharat-finance-platform>

AI Speech Persona Builder

2025 • Full-Stack: FastAPI • Next.js • Gemini API • WebSocket • Real-time Processing

Intelligent speech-coaching platform analyzing delivery metrics (pace, vocabulary, tone) to deliver real-time, personalized feedback.

Demonstrates full-stack expertise: real-time audio processing, WebSocket communication, AI integration, and responsive UI.

Architecture & Technical Highlights:

Backend: FastAPI with async handlers for real-time WebSocket communication

Audio Processing: Stream-based audio capture, encoding, and AI model integration
AI Integration: Gemini API for speech analysis, metrics extraction, and personalized feedback generation
Frontend: Next.js UI with WebSocket client for real-time feedback visualization
GitHub Repository: <https://github.com/pavitagrawal/ai-speech-coach>

INTERNSHIP & TRAINING PROJECTS

ML-Driven Fraud Detection System Integration

TransOrg Analytics • May 2025 – Jul 2025

Contributed to production integration of deep-learning fraud detection engine. Designed feature engineering pipelines, performed model validation, and created data processing workflows. Collaborated with ML engineers and backend teams on system architecture and performance optimization.

Responsibilities & Technical Contribution:

Analyzed transaction datasets and engineered features for model training and validation
Built SQL pipelines for ETL processes and data preparation
Validated model predictions, conducted performance benchmarking, and ROC-AUC analysis
Documented ML pipeline architecture and created runbooks for production deployment
Collaborated with data science team on model improvements and optimization strategies

Retail Discount Optimization & Analytics Dashboard

TransOrg Analytics • May 2025 – Jul 2025

Conducted comprehensive product-level sales analysis to optimize retail discount strategies. Built analytical frameworks, statistical models, and BI dashboards. Identified margin elasticity patterns and quantified revenue impact of promotional initiatives.

Responsibilities & Technical Contribution:

Exploratory Data Analysis (EDA) on sales, pricing, and discount datasets using Python
SQL query optimization for large-scale sales data aggregation and analysis
Built Power BI dashboards with interactive visualizations for stakeholder reporting
Statistical modeling to calculate margin elasticity and identify optimal discount thresholds
Delivered actionable insights and strategic recommendations for pricing optimization

Credit Data EDA & Financial Analytics Pipeline

IIT Guwahati Industrial Data Science Internship (1stop.ai)

Conducted comprehensive exploratory data analysis on credit and financial datasets. Uncovered trends, identified anomalies, and built interactive dashboards for stakeholder decision-making.

Technical Implementation:

Python-based EDA with pandas, NumPy, Matplotlib, and Seaborn
Statistical analysis: correlation studies, distribution analysis, outlier detection
Data visualization with interactive dashboards for stakeholder reporting
Documentation of findings and business insights for actionable recommendations
GitHub Repository: <https://github.com/pavitagrawal/credit-data-eda>

Heart Failure Mortality Prediction Model

IIT Guwahati Industrial Data Science Internship (1stop.ai)

Developed machine learning models to predict heart failure mortality risk from clinical datasets. Applied advanced statistical methods and feature selection to identify key prognostic indicators. Achieved high predictive accuracy and interpretability.

Technical Implementation:

Models: Logistic Regression, Gradient Boosting, ensemble methods for mortality prediction
Feature Engineering: Domain-driven feature selection based on clinical knowledge
Validation: Cross-validation, ROC-AUC analysis, confusion matrix analysis, performance metrics
Interpretability: Risk stratification and clinical decision support framework
GitHub Repository: <https://github.com/pavitagrawal/heart-failure-prediction>

Hate Speech Detection NLP System

IIT Guwahati Industrial Data Science Internship (1stop.ai)

Built NLP-based text classification system to detect hate speech and toxic content. Applied advanced preprocessing techniques, feature engineering, and ensemble machine learning methods to achieve high classification accuracy.

Technical Implementation:

Preprocessing: Tokenization, lemmatization, stopword removal, and vectorization
Feature Extraction: TF-IDF vectors, Word2Vec embeddings, GloVe representations
Classification Models: Logistic Regression, SVM, Random Forest, Gradient Boosting
Evaluation: Precision, recall, F1-score, confusion matrices, cross-validation analysis
GitHub Repository: <https://github.com/pavitagrawal/hate-speech-detection>

ACHIEVEMENTS & COMPETITIONS

■ HackverseX 2026

Winner Certificate – Recognized for innovative AI/ML solution and full-stack development excellence

■ National Geo AI Hackathon, IIT Bombay Techfest 2025

Finalist Recognition for Hydro-DTM System – High-precision terrain classification achieving 95%+ accuracy with production-grade ML pipeline

■ AWS ImpactX Hackathon, IIT Bombay Techfest 2025

Built Bharat Finance Platform – Full-stack FinTech application with multilingual interface and real-time visualizations on AWS infrastructure

■ **ET AI Hackathon 2.0**

Team Project: Mnemos Platform – Advanced agentic AI system for industrial knowledge management with hybrid retrieval

■ **Hackefest 2026**

Participation – Competed in full-stack development and system design challenges

■ **YouthCodeXAI 2026**

Certificate – Active participation in AI learning community and hands-on workshops

PROFESSIONAL CERTIFICATIONS

AWS Certified Cloud Practitioner

AWS cloud architecture, services, best practices, and deployment strategies

Google AI Certification (Coursera)

Credential ID: 5QN2YVUFVUD6 | Machine Learning fundamentals and AI application design

Deloitte Control Assurance & IT Audit

Enterprise control assessment, cybersecurity validation, and compliance auditing

TransOrg Analytics Data Science Training

Comprehensive industry training: SQL optimization, Python data manipulation, Power BI, Tableau, statistical modeling

IIT Guwahati Industrial Data Science (1stop.ai)

Advanced ML and statistical modeling across healthcare, finance, and social datasets

NXT Wave AI Workshop Masterclass

Advanced AI/ML concepts, practical applications, and industry best practices

Data Vista 2024 Workshop Series

Generative AI, Retrieval-Augmented Generation (RAG), and advanced BI tool mastery

PORTFOLIO & RESOURCES

■ Portfolio Website: <https://pavitagrawal.dev>

■ **GitHub Profile:** <https://github.com/pavitagrawal>

■ LinkedIn Profile: <https://linkedin.com/in/pavit-agrawal-736572283>

■ Email: pavitagrawal@gmail.com

■ Phone: +91 9560075025

Note: This comprehensive CV highlights full-stack software engineering expertise, from backend architecture and database design to frontend development and cloud deployment. For detailed project case studies, code walkthroughs, and additional materials, please visit the GitHub repositories and portfolio website listed above.

Updated: September 2026